

1. How does shor's algorithm threaten RSA and ECC and what are the consequences?

Ans: Shor's algorithm is a quantum algorithm that can solve the two math problems RSA and ECC depend on factoring large numbers (RSA) and the discrete logarithm problem (ECC) - much faster than any known classical algorithm. If a large, error-connected quantum computer exists, it could recover private keys from public keys and break the security of that systems that rely on RSA and ECC.

What shor does:

It is a quantum algorithm that can efficiently factor large integers and compute discrete logarithms.

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• why that matters: it implies that we can't just ignore the problem.

1. RSA security: hardness of factoring large numbers. Shor's $\Rightarrow$ private RSA keys can be recovered from public key.

2. ECC security: hardness of elliptic-curve discrete logs. Shor's $\Rightarrow$ ECC private keys can be recovered from public keys.

Consequences for digital infrastructure:

Decrypted past/future communications, forged digital signatures, broken TLS/HTTPS,

corrupted software updates, compromised

banking and government systems, and

"harvest-now, decrypt-later" attacks.

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- Current status and defence:

Large fault-tolerant quantum machines able to do this do not yet exist; but migration to post-quantum algorithms and hybrid schemes should stand now.

2. Discuss the role of QKD in future cryptographic systems. How does it differ from classical public key encryption?

• QKD: Uses quantum properties to create shared symmetric key while allowing detection of eavesdroppers.

- Differences from public key:

1. QKD's security is based on physics. Classical PKI (RSA/ ECC) is based on computational hardness.
2. QKD gives you secure key. You still need authentication.

3. QKD requires special hardware, is costly, has distance/throughput limits - not a universal drop-in replacement for PKI.

When to use: High security links where long term secrecy matters otherwise

Post-quantum classical algorithms are more practical for general use.

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QKD is based on computational hardness

QKD gives for secure key. For still need authentication

3.1- what are the main difference between lattice-based cryptography and traditional number-theoretic approaches like RSA, particularly in the context of quantum resistance?

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Answer

• RSA/ECC rely on factoring or discrete-log problems - broken by Shor's algorithm (Quantum). not secure

• Lattice-based crypto (e.g. LWE, NTRU, schemes used in Kyber/Dilithium) relies on geometric problems in high-dimensional lattices (like shortest vector problem). No efficient

Quantum algorithm is known for these -
 So they are strong PQC (Post-quantum
 Cryptography) candidates.

• Trade-offs:

Lattice schemes often have longer keys / ciphertexts, but are fast and have solid theoretical reductions. They are leading practical approach for post-quantum migration.

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4. Develop a python based PRNG that uses the current system time and a custom seed value. Write complete program and corresponding output:

Source code:

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```
import time, random
```

```
def prng_with_seed(custom_seed, n=10):
```

```
    system_time = int(time.time())
```

```
    combined_seed = system_time ^ custom_seed
```

```
    random.seed(combined_seed)
```

```
    seq = [random.randint(0, 100) for _ in range(n)]
```

```
    return system_time, combined_seed, seq
```

```
if __name__ == '__main__':
    system_time, combined_seed, seq = prng_with_seed(12345, 10)
```

P.T.O

now both ways based on the following code .
 Custom_seed = 12345
 System_time, combined_seed, seq = prng - with
 seed (custom_seed, 10)
 : kugtwo prbnoqesnoo

```
Print ("System time :", system_time)
Print ("Custom seed:", custom_seed)
Print ("Combined seed:", combined_seed)
Print ("Pseudo-Random numbers (0-100):", seq)
```

Output: (yaml)

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System time: 1723468907

Custom time: 12345

Combined seed: 1723480194

Pseudo Random numbers (0-100): [14, 67, 40, 96,
89, 53, 65, 6, 28, 100]

5. Explain the Sieve of Eratosthenes algorithm and use it to find all prime numbers less than 50.
How does its time complexity compare to trial division?

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Ans: not (n/p) for all p | n. This is how the sieve works!

1. Write the number 2 to 49.
2. Start at 2 and cross out every multiple of 2 (4, 6, 8, ...)
3. Next unmarked is 3. Cross out every multiple of 3 (6, 9, 12, ...).
4. Next unmarked is 5, cross multiple of 5
5. Continue until you reach numbers $> \sqrt{50}$ ($\sqrt{50} \approx 7.07$), so you need only do 2, 3, 5, 7
6. The remaining unmarked numbers are primes.

Primes less than 50:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

Time complexity:

• Sieve: about $O(n \log \log n)$ for primes up to n , very efficient in range.

• Trial division: Testing each number up to n by dividing up to $\sqrt{n}$ leads to about $O(n\sqrt{n})$ overall.

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6. State and explain the necessary and sufficient conditions for a composite number to be a Carmichael number. Then verify whether the number $n=561$ and $n=1105$ and $n=1729$ are Carmichael number?

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Ans:

A composite number is a Carmichael number if

1. n is square-free and
2. for every prime p dividing n , $(p-1)$ divides $(n-1)$

for, $n=561$

- factor $561 = 3 \times 11 \times 17$ (all distinct - squarefree)
- $n-1 = 560$. Check $(p-1) | 560$:
 - $3-1 = 2 \rightarrow 2 | 560$ ✓
 - $11-1 = 10 \rightarrow 10 | 560$ ✓
 - $17-1 = 16 \rightarrow 16 | 560$ ✓

561 is Carmichael

for $n=1105$

$$1105 = 5 \times 13 \times 17 \text{ (square free)}, (n-1) = 1104$$

$$4 \mid 1104 \checkmark, 12 \mid 1104 \checkmark, 16 \mid 1104 \checkmark$$

1105 is carmichael.

Ans

$$1729 = 7 \times 13 \times 19 \text{ (square free)}, (n-1) = 1728$$

$$6 \mid 1728 \checkmark$$

$$12 \mid 1728 \checkmark$$

$$18 \mid 1728 \checkmark$$

1729 is carmichael.

$$(n-1) = 1728$$

$$3-1 = 2 \Rightarrow 5 \mid 1728 \checkmark$$

$$11-1 = 10 \Rightarrow 10 \mid 1728 \checkmark$$

7. Determine whether the following are valid algebraic structures and justify your answer:

• Is the set $\mathbb{Z}_{11}$ with operations $(+, \cdot)$ a ring?

• Are the sets $(\mathbb{Z}_{37}, +)$ and $(\mathbb{Z}_{35}, \times)$ abelian groups?

Ans:

a) Is $\mathbb{Z}_{11}$ with $(+, \cdot)$ ring?

Yes, integers mod 11 with addition and multiplication mod 11 satisfy ring axioms. Since 11 is prime, every non-zero element has multiplicative inverse $\rightarrow$ it's actually a field.

b) Are $(\mathbb{Z}_{37}, +)$ and $(\mathbb{Z}_{35}, \times)$ Abelian groups?

$(\mathbb{Z}_{37}, +)$: Yes, addition modulo 37 is associative, commutative, has identity 0, and every element has an additive inverse. So, it's an abelian group.

• $(\mathbb{Z}_{35}, \times)$: No, the set of all residues modulo 35 under multiplication is not a group because 0 has no inverse and elements not co-prime to 35 (like 5) have no inverse. However, the set of units modulo 35, written $\mathbb{Z}_{35}^*$, is an abelian group under multiplication.

Question-8: what is the remainder when -52 is reduced modulo 31?

Answer

1. Find an equivalent positive remainder:
Add 31 enough times until non-negative
2. $-52 + 62 = 10$ (since $62 = 2 \times 31$)
So $-52 \equiv 10 \pmod{31}$

9. Determine the multiplicative Inverse of $7 \pmod{26}$ if it exists.

(Note) $21 \equiv -5 \pmod{26}$; substitute -5 for 21 .

Ans:

We will solve $7x \equiv 1 \pmod{26}$ to find x

1. Run Euclidean algorithm on $(26, 7)$:

$$\Rightarrow 26 = 7 \cdot 3 + 5$$

$$\Rightarrow 7 = 5 \cdot 1 + 2$$

$$\Rightarrow 5 = 2 \cdot 2 + 1$$

$$\Rightarrow 2 = 1 \cdot 2 + 0$$

2. Back substitute to express 1 as combination of 7 and 26:

$$\Rightarrow 1 = 5 - 2 \cdot 2$$

$$\Rightarrow \text{But } 2 = 7 - 5 \cdot 1 \rightarrow 1 = 5 - 2(7 - 5) = 3 \cdot 5 - 2 \cdot 7$$

$\Rightarrow 3 \cdot 5 - 2 \cdot 7 = 1$. Hence the coefficient of 7 is -2 , giving inverse $x \equiv -2 \pmod{26}$

2. Answer

4. Convert to standard residue: $-11 \equiv 15 \pmod{26}$

5. Multiplicative inverse of 7 modulo 26 is 15

(check: $7 \times 15 = 105 \equiv 1 \pmod{26}$)

Answer

40. Evaluate $(-8 \times 5) \pmod{17}$ and explain

how to simplify negative modular multiplication

Ans: Method-1: (replace neg by pos residue)

$$-8 \equiv 9 \pmod{17} \quad \text{because } 17 - 8 = 9$$

then $9 \times 5 = 45$. Reduce $45 \pmod{17} = 45 - 2 \times 17 = 45 - 34 = 11$

$11 \equiv x \pmod{17}$

method 2 (direct):

$$\Rightarrow -8 \times 5 = -40 \dots \text{Add } 3 \times 17 = 51; -40 + 51 = 11$$

(288 bonr) + 11 = 11

$$(-8 \times 5) \bmod 17 = 11 \quad (\text{Answer})$$

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multinomial

11. State and prove Bezout's theorem. Use it to find the multiplicative inverse of 97 modulo 385.

Ans: For any integers a, b there exist integers x, y such that $ax + by = \gcd(a, b)$.
if $\gcd(a, b) = 1$ then there are integers x, y with $ax + by = 1$. The coefficient x gives the most modular inverse of a modulo b .

modulo the order. (1000) 5 bottom

Finding the inverse. $97n \equiv 1 \pmod{385}$

We want n with $97n \equiv 1 \pmod{385}$

find integers n, y with $97n + 385y = 1$.

1. Euclidean Algorithm.

$$\Rightarrow 385 = 97 \cdot 3 + 94$$

$$\Rightarrow 97 = 94 \cdot 1 + 3$$

$$\Rightarrow 94 = 3 \cdot 31 + 1$$

$$\Rightarrow 3 = 1 \cdot 3 + 0$$

2. Back substitution to express 1:

$$\Rightarrow 1 = 94 - 3 \cdot 31$$

$$\Rightarrow \text{But } 3 = 97 - 94 \Rightarrow 1 = 94 - 31(97 - 94) = 32 \cdot 94 - 31 \cdot 97$$

$$\Rightarrow \text{And } 94 = 385 - 97 \cdot 3 \Rightarrow 1 = 32(385 - 97 \cdot 3) - 31 \cdot 97$$

$$\Rightarrow 1 = 32 \cdot 385 - 96 \cdot 97 - 31 \cdot 97$$

$$\Rightarrow \text{Combine terms for } 97: 1 = 32 \cdot 385 - 127 \cdot 97$$

$$\Rightarrow 1 = 32 \cdot 385 - 127 \cdot 97$$

• Combine 43 terms : $7+60 = 67$. So $1 = 67 \cdot 43$
— 12.240

3. Thus $67 \cdot 43 \equiv 1 \pmod{240}$

$X = 67$ (Since $43 \times 67 = 2881 = 240 \times 12 + 1$).

Ans

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12. Using Bezout's identity, prove that the equation $ax + by = \gcd(a, b)$ has integer solution. Find x such that $43x \equiv 1 \pmod{240}$.

Ans: Bezout identity: For integers a, b there exist integers x, y with $ax + by = \gcd(a, b)$. This guarantees a solution exists; if $\gcd(a, b) = 1$ then the equation $ax + by = 1$ has integer solution and x is the inverse of $a \pmod{b}$.

Find x with $43x \equiv 1 \pmod{240}$:

1. Euclidean algorithm on $(240, 43)$:

- $240 = 43 \cdot 5 + 25$ (remainder 25)
- $43 = 25 \cdot 1 + 18$
- $25 = 18 \cdot 1 + 7$
- $18 = 7 \cdot 2 + 4$
- $7 = 4 \cdot 1 + 3$
- $4 = 3 \cdot 1 + 1$
- $3 = 1 \cdot 3 + 0$

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2. Back substitution to express 1:

- $1 = 4 - 3 \cdot 1$
- $3 = 7 - 4 \cdot 1 \Rightarrow 1 = 4 - (7 - 4) = 2 \cdot 4 - 7$
- $4 = 18 - 7 \cdot 2 \Rightarrow 1 = 2(18 - 7 \cdot 2) - 7 = 2 \cdot 18 - 5 \cdot 7$
- $7 = 25 - 18 \Rightarrow 1 = 2 \cdot 18 - 5(25 - 18) = 7 \cdot 18 - 5 \cdot 25$
- $18 = 43 - 25 \Rightarrow 1 = 7(43 - 25) - 5 \cdot 25 = 7 \cdot 43 - 12 \cdot 25$
- $25 = 240 - 43 \cdot 5 \Rightarrow 1 = 7 \cdot 43 - 12(240 - 43 \cdot 5) = 7 \cdot 43 - 12 \cdot 240 + 60 \cdot 43$
- combine 43 terms: $7 + 60 = 67$. So $1 = 67 \cdot 43 - 12 \cdot 240$

3. Thus $67 \cdot 43 \equiv 1 \pmod{240}$.

Ans: $x = 67$ (since $43 \times 67 = 2881 = 240 \times 12 + 1$).

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13. State and prove the Chinese Remainder theorem.

Then solve the following system of congruences

$$x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 2 \pmod{7}$$

- FLT: If p is prime and $\gcd(a, p) = 1$ then $a^{p-1} \equiv 1 \pmod{p}$. One proof: the non-zero residues mod p form a multiplicative group of size $p-1$; hence a^{p-1} is the identity.
- Use for primality: If for some base a with $\gcd(a, n) = 1$ we find $a^{n-1} \not\equiv 1 \pmod{n}$, then n is composite. But the converse fails, so it's a compositeness test, not a full primality test.
- 561: It is not prime. It's a combined num. $a^{560} \equiv 1 \pmod{561}$ for all $\gcd(a, 561) = 1$. So FLT can be fooled by 561.
- Compute $5^{123} \pmod{175}$: since $175 = 25 \cdot 7$ and $5^{123} \equiv 0 \pmod{25}$. mod 7: $\varphi(7) = 6 \Rightarrow 5^{123} = 5^3 = 125$
- Solve $x \equiv 0 \pmod{25}$, $x \equiv 6 \pmod{7}$: write $x = 25k$
so $25k \equiv 6 \pmod{7} \Rightarrow 4k \equiv 6 \pmod{7} \Rightarrow k \equiv 5 \pmod{7}$
- Take $k = 5 \Rightarrow x = 25 \cdot 5 = 125$

Ans: $5^{123} \equiv 125 \pmod{175}$

14. State and prove the Chinese Remainder theorem. Then solve the following system of congruences:

$$x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 2 \pmod{7}$$

Ans:

- CRT: For pairwise coprime $n_1, \dots, n_k$ and any residues a_i , the system $x \equiv a_i \pmod{n_i}$ has a unique solution modulo $N = \prod n_i$.
construction: let $N_i = N/n_i$ and choose y_i with $N_i y_i \equiv 1 \pmod{n_i}$; then $x = \sum a_i N_i y_i$ works
- Here $n_1 = 3, n_2 = 5, n_3 = 7$ (pairwise coprime), $N = 105$.
- Compute $N_1 = 35$ with inverse $y_1 = 2 \pmod{3}$; $N_2 = 21$ with $y_2 = 1 \pmod{5}$; $N_3 = 15$ with $y_3 = 1 \pmod{7}$.
- $x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 140 + 63 + 30 = 233$
- Reduce mod 105: $233 - 2 \cdot 105 = 23$.
- Answer: $x \equiv 23 \pmod{105}$.

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15. Briefly explain the CIA triad in information security. How does it component to building a secure system?

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Ans:

- Confidentiality : Only authorized entities may read data. Achieved via access control, encryption, strong authentication, and least-privilege policies. It prevents disclosure/leakage.
- Integrity : Data is accurate and unaltered. Achieved with checksums, hashes, digital signature, secure audit logs, and control of write privileges. Detects and deters tampering.
- Availability : Systems and data remain accessible when needed. Achieved using redundancy, failover, load-balancing, backups, patching and DoS protections.
- A secure system balances all three
- Governance (risk assessment, policies) and technical controls map each business asset to specific CIA requirements.

16. How does steganography differ from in the context of information security and what are common techniques used for hiding data in digital media? Mirza Zisun

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Ans:

- Cryptography hides the content (plaintext $\rightarrow$ ciphertext) but signals that a secret exists.
- Steganography hides the content, the existence of the message inside a benign carrier, aiming not to rise suspicion.
- Typical digital techniques: image LSB embedding, palette, modification and transformation-domain methods (DCT/DWT coefficients in JPEG/Video).
- Audio: echo hiding, spread-spectrum, phase coding. Video: inter-frame embedding; text: whitespace or font perturbations.
- Best practice: encrypt first, then hide; if discovered, the payload remains unintelligible.
- Trade-offs: Capacity vs. imperceptibility vs. robustness to compression/edits; steganalysis targets statistical anomalies.

17. What are the key differences between phishing, malware, and denial-of-service (DoS) attacks in terms of their method and impact on system security?

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Ans:

- **Phishing:** Social-engineering to trick users into giving credentials, or running payloads. Impact: account takeover, initial footholds, data theft, mitigate with awareness training, email filters, DMARC, MFA.
- **Malware:** Malicious software that executes on hosts to exfiltrate, encrypt or disrupt, Mitigate via EDR/AV, patching least privilege, application allow-listing.
- **DoS/DDoS:** Overwhelm resources to break availability. Mitigate with scrubbing centers, rate-limit, anycast/CDN, autoscaling.
- **Differences:** phishing targets people; malware targets endpoints/data; DoS targets service uptimes. Indicate may chain (phish → malware → DDOS as diversion)

18. Explain how legal frameworks such as the General Data Protection Regulation (GDPR) help mitigate attacks and protect user privacy.

Ans:

- Establish principles that directly reduce the data an attacker could steal.
- Requires appropriate technical and organizational measure and privacy by design/default in system.
- Mandates breach notification, creating strong incentives for monitoring and rapid response.
- Grants data-subject rights that curb uncontrolled retention and shadow data stores.
- Enforces with significant fines, pushing executive-level accountability and budget for security programs.
- While not a technical control, GDPR aligns governance and engineering to shrink attack surface and breach impact.

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19. Explain the basic working of the DES algorithm using a simple 64-bit plaintext block and a 56-bit-key. Show how the initial permutation, round functions and final permutation contribute the encryption process.

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Ans:

- DES applies an initial permutation (IP) to a 64 bit plaintext, then process 16 Feistel rounds over halves L_i, R_i .
- Each round : $L_i = R_{i-1} + F(R_{i-1}, K_i) = L_{i-1} \oplus F(R_{i-1}, K_i)$. The round function f : expand R_i from 32 $\rightarrow$ 48 bits (E-box), XOR with subkey K_i , pass 8 groups through s-boxes (nonlinear 6 $\rightarrow$ 4 mapping) then permute (P-box).
- The Feistel structure ensures decryption uses the same steps with keys in reverse order.
- After 16 rounds, swap halves and apply the inverse IP to get ciphertext.
- IP/final perm permutations are fixed bit shuffles aiding diffusion across s-boxes; security hinges on s-boxes and key schedule.

20. In the DES algorithm, a 64 bit plaintext block is divided into two 32-bit halves L_0 and $R_0 = 0xFOFOFOFO$ and round key $K_1 = 0xFOFOFOF$, compute the output of the first round's function $f(R_0, K_1)$ assuming XOR operation only. Then, find $L_1 = R_0$ and $R_1 = L_0 \oplus f(R_0, K_1)$, where $L_0 = 0xAAAAAAAA$.

Ans:

- Given $R_0 = 0xFOFOFOFO$, $K_1 = 0xFOFOFOF$, and $L_0 = 0xAAAAAAAA$
- With the stated simplification, $f(R_0, K_1) = R_0 \oplus K_1$
- Compute per byte: $0xFO \oplus 0xOF = 0xFF$; repeat
 $\rightarrow f = 0xFFFFFFFF$
- Then $L_1 = R_0 = 0xFOFOFOFO$
- $R_1 = L_0 \oplus f = 0xAAAAAAAA \oplus 0xFFFFFFFF = 0x55555555$
- Answer: $f = 0xFFFFFFFF$, $L_1 = 0xFOFOFOFO$, $R_1 = 0x55555555$

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21. Given the input word:

$[0x23, 0xA7, 0x4E, 0x19]$

use the following partial AES S-box to perform SubBytes transformation. Provide the resulting output word.

Row $\downarrow$ \ Col $\rightarrow$ 3 4 7 9 A C

6 D . . C 6 . .

0 4

: A 1 2 E

: . 6 3 . 0 2 .

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Ans: SubBytes applies the AES S-box independently to each byte (row = high nibble, column = low nibble).

• Using the standard S-box table:

• $0x23 \Rightarrow S[2,3] = 0x26$

• $0xA7 \Rightarrow S[A,7] = 0x5C$

• $0x4E \Rightarrow S[4,E] = 0x29$

• $0x19 \Rightarrow S[1,9] = 0xD4$

• Output word: $[0x26, 0x5C, 0x29, 0xD4]$

• (Process recap: convert each byte to row/column nibbles, look up S-box value, replace, keep byte order unchanged.)

22. In AES encryption, apply the AddRoundKey step only. Given :

- Input word : $[0x1A, 0x2B, 0x3C, 0x4D]$
- Round key word : $[0x55, 0x66, 0x77, 0x88]$

perform XOR between the input word and the round key and write the resulting output word.

Ans: Input word : $[0x1A, 0x2B, 0x3C, 0x4D]$

Round key : $[0x55, 0x66, 0x77, 0x88]$

AddRoundKey performs a bitwise XOR between state the round key. Compute per byte :

- $0x1A \oplus 0x55 = 0x4F$ ($0x1A = 0011010, 0x55 = 01010101 \rightarrow \text{XOR} = 01001111$)
- $0x2B \oplus 0x66 = 0x4D$
- $0x3C \oplus 0x77 = 0x4B$
- $0x4D \oplus 0x88 = 0xC5$

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So the resulting output word after AddRoundKey is $[0x4F, 0x4D, 0x4B, 0xC5]$. This steps mixes the round key into the state and is reversible by XORing with the same round key.

23. Show how MixColumns uses the following fixed matrix over $GF(2^8)$:

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$

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to transform an input column. Use an example column with values $[0x01, 0x02, 0x03, 0x04]$.

Ans: Let column $x = [x_0, x_1, x_2, x_3] = [0x01, 0x02, 0x03, 0x04]$. Multiplication by 02 is xtime (left shift with conditional XOR $0x18$), and $03 = 02 \oplus 01$. Compute each output byte y_i :

- $y_0 = 2 \cdot 0x01 \oplus 3 \cdot 0x02 \oplus 1 \cdot 0x03 \oplus 1 \cdot 0x04$
 $= 0x02 \oplus (0x04 \oplus 0x02) \oplus 0x03 \oplus 0x04 = 0x03$
- $y_1 = 1 \cdot 0x01 \oplus 2 \cdot 0x02 \oplus 3 \cdot 0x03 \oplus 1 \cdot 0x04$
 $= 0x01 \oplus 0x04 \oplus (0x06 \oplus 0x03) \oplus 0x04$
- $y_2 = 1 \cdot 0x01 \oplus 1 \cdot 0x02 \oplus 2 \cdot 0x03 \oplus 3 \cdot 0x04$
 $= 0x01 \oplus 0x02 \oplus 0x06 \oplus (0x03 \oplus 0x04) = 0x05$
- $y_3 = 3 \cdot 0x01 \oplus 1 \cdot 0x02 \oplus 1 \cdot 0x03 \oplus 2 \cdot 0x04$
 $= (0x02 \oplus 0x02 \oplus 0x03 \oplus 0x03) \oplus 0x04 = 0x0A$

So the column is $[0x03, 0x04, 0x05, 0x0A]$. MixColumns provides diffusion by mixing bytes of each column using $GF(2^8)$ arithmetic.

24. Describe how AES-OFB mode works. How does it ensure synchronization between encryption and decryption streams?

Ans: OFB turns a block cipher into a stream cipher by repeatedly encrypting an initial vector to produce keystream blocks. Procedure: start with IV; compute $O_i = E_K(IV)$, then $O_{i+1} = E_K(O_i)$. XOR each keystream block O_i with plaintext block P_i to produce ciphertext C_i . Decryption regenerates the identical keystream and XORs with ciphertext to recover plaintext. Synchronization is deterministic: both ends must use the same IV and process blocks in the same order to produce identical O_i . OFB is resilient to lost ciphertext blocks, but both sides must maintain the same position in the keystream; an IV reuse is catastrophic (keystream reuse leads to plaintext XOR leakage).

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25. Which AES modes cause error propagation during decryption, and how does this affect the integrity of the decrypted message? Illustrate with CBC and CFB modes.

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Ans: Error propagation depends on feedback and XOR structure.

- CBC: Decryption computes $P_i = D_K(C_i) \oplus C_{i-1}$. A single-bit error in C_i corrupts the entire decrypted block P_i and also flips the same bit positions in P_{i+1} . So one corrupted ciphertext block affects two plaintext blocks - wide propagation and integrity damage.
- CFB: Decryption uses the previous ciphertext encrypted and XORed with current ciphertext. A bit error in C_i will typically cause bit errors in the corresponding decrypted plaintext bytes for that block, but the corruption shifts out according to the feedback size and doesn't usually many subsequent blocks; error propagation is limited.

Implications: modes with stronger propagation make tampering damaging; use authenticated encryption or a MAC to ensure integrity and detect tampering.

26. Which AES mode would you recommend for encrypting large files with parallel processing and why? Justify your choice between ECB, CBC and CTR.

Ans: CTR mode is recommended for encrypting large files where parallelism matters. CTR mode generates keystream blocks by encrypting independent counter values E_k ; each block's keystream can be computed independently so encryption and decryption are trivially parallelizable and support random-access to any block. ECB is parallel but insecure because identical plaintext blocks map to identical ciphertext blocks. CBC requires sequential chaining for encryption (each ciphertext depends on previous on previous ciphertext), so encryption is not parallelizable (decryption can be parallel). CTR provides strong parallel performance, random access, and is secure if nonces/counters are never reused.

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27. Given a message "A" represented as $M = 1$. Encrypt it using public key $e = 5, n = 14$. What is the ciphertext? Then decrypt using private key $d = 11$.

Ans: RSA encryption: $c \equiv m^e \pmod{n}$. With $m = 1$, any positive power remains 1: $1^5 \equiv 1 \pmod{14}$. So ciphertext $c = 1$. Decryption: $m \equiv c^d \pmod{n} = 1^{11} \equiv 1$. Thus encryption and decryption both yield 1. Note: this is a toy example - $n = 14$ is small and composite (14 = 2 * 7), so this RSA setup is insecure in practice; nevertheless the arithmetic shows the process and result.

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28. Given: Message hash: $H(m) = 5$, RSA Private key: $d = 3$, $n = 33$. Generate the digital Signature.

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Ans: A simple RSA signature is $S \equiv H(m)^d \pmod{n}$. compute $5^3 = 125$. Reduce modulo 33: $125 - 3 \cdot 33 = 125 - 99 = 26$. So the signature is $S = 26$. Verification (not asked but standard) would check that $S^e \pmod{n} = H(m)$ using the public exponent e .

29. Aleya and Badal are using the Diffie-Hellman key exchange protocol. They agree on the following public values: Prime modulus: $p=17$, Base (generator): $g=3$, Aleya chooses a private key, $a=4$, Badal chooses a private key, $b=5$. Compute public key of Aleya and Badal.

Ans: Each party computes public key $A = g^a \mod p$, $B = g^b \mod p$. compute:

- Aleya's public $A = 3^4 = 81$. Reduce mod 17: $81 - 4 \cdot 17 = 81 - 68 = 13$. So $A = 13$.
- Badal's public $B = 3^5 = 243$. Reduce mod 17: $19 = 238 \rightarrow 243 - 238 = 5$. So $B = 5$.

Shared secret: $s = B^a \mod p = 5^4 = 625 \mod 17$. $17 \cdot 36 = 612 \rightarrow 625 - 612 = 13$. Alternatively

$A^b = 13^5 \mod 17$ reduces to same value.

So the shared secret is 13. This secret is used to derive symmetric keys.

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30. A simple hash function $H(x)$ is defined as the sum of ASCII values of the characters in a message modulo 100:

$$H(x) = (\sum \text{ASCII of characters in } x) \bmod 100$$

compute the hash value of the message "AB" and "BA" using this function. Do the two messages produce the same hash? What does this imply about collision resistance in weak hash functions?

Ans: Hash: $H(x) = (\sum \text{ASCII}) \bmod 100$

"AB" : $65 + 66 = 131 \rightarrow 131 \bmod 100 = 31$

"BA" : $66 + 65 = 131 \rightarrow 31$

Both have same hash $\rightarrow$ collision occurs.

Shows weak collision resistance in such simple functions.

Small changes in input can give same output.

Not secure for cryptographic use.

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31. In a secure message system, a simple message authentication code (MAC) is computed using modular addition:

$$\text{MAC} = (\text{message} + \text{Secret key}) \bmod 17$$

Given: Message: 15, Secret key: 7. Compute the MAC for the message. Suppose an attacker changes the message to 10 but doesn't know the key. Can they forge the correct MAC easily? Explain briefly.

Ans:

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$$\text{MAC} = (\text{Message} + \text{Secret key}) \bmod 17$$

$$\text{Given: } (15 + 7) \bmod 17 = 22 \bmod 17 = 5$$

$$\text{If message changes to 10} \rightarrow \text{MAC} = (10 + 7) \bmod 17 = 0$$

Key can be found from one valid MAC: $k = (\text{MAC} - \text{message}) \bmod 17$.

Attacker can forge MAC easily.

Insecure scheme due to simple linear relation

32. Explain the steps involved in the TLS handshake process. How are symmetric keys established securely using asymmetric cryptography during the handshake?

Ans: TLS handshake:

1. ClientHello → algorithms, random number.
2. ServerHello → Chosen algorithms, cert, key share.
3. Client verifies certificate.
4. Exchange key shares via DH/ECDH.
5. Derive symmetric session keys.
6. Exchange finished messages.
7. Encrypted communication starts.

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33. Explain the layered architecture (protocol stack) of SSH. Briefly describe the roles of each layer.

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Ans: SSH protocol stack:

- Transport layer : encryption, integrity, key exchange.
- User authentication : verifies client identity.
- Connection layer : manages channels over secure link.

Provides confidentiality, integrity, authentication.

Each layer builds on previous one.

Supports shell, exec, SFTP, port forwarding

39. Explain the steps involved in the TLS handshake process.

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Ans: TLS handshake brief:

1. ClientHello (cipher, random)
2. ServerHello (chosen, cipher, cert, key share).
3. Client sends key share.
4. DH/ECDH computes shared secret.
5. Derive Finished message.
6. Send Finished message.
7. Switch to encrypted session.

35. What is the general form of an elliptic curve equation over a finite field and why is it used in cryptography?

Ans: Elliptic curve over finite field:

$$y^2 = x^3 + ax + b \pmod{p}$$

Condition: $4a^3 + 27b^2 \neq 0$ to avoid singularities.

Used because ECDLP is hard.

Small keys give strong security.

Efficient in bandwidth and computation.

Ideal for constrained environments.

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Q6. How does ECC achieve the same level of security as RSA with a smaller key size? Briefly explain.

Ans: ECC security:

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ECDLP harder than factoring (RSA).

RSA 3072-bit $\approx$ ECC 256-bit security.

Smaller keys $\rightarrow$ less storage, bandwidth.

Faster computations, less power use.

Good for mobile/IoT.

Same security with reduced resources.

37. Given the elliptic curve $y^2 = x^3 + 2x + 3 \pmod{97}$, determine whether the point $P = (3, 6)$ lies on the curve.

Ans: Curve: $y^2 = x^3 + 2x + 3 \pmod{97}$, $P = (3, 6)$

LHS: $6^2 = 36$

RHS: $27 + 6 + 3 = 36$

$36 \pmod{97} = 36$ for both.

Point lies on curve.

Valid for ECC operations.

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38. Given public key $(p=23, g=5, h=8)$ and message $m=10$, compute the ElGamal ciphertext using random $k=6$.

Ans: ElGamal: $p=23, g=5, h=8, m=10, k=6$

$$C_1 = 5^6 \bmod 23 = 8$$

$$C_2 = 10 \times 8^6 \bmod 23 = 10 \times 13 \bmod 23 = 15$$

Ciphertext: $(8, 15)$

Random k ensures security.

Do not reuse k .

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39. Explain how lightweight cryptography is important for securing IoT devices. Give one example of lightweight encryption algorithm used in IoT.

Ans: Lightweight cryptography is crucial for IoT because devices often have low CPU, RAM, power and bandwidth. Standard algorithms like AES-256 or RSA-3072 can be too heavy for such devices. Lightweight crypto is designed to be efficient in code size, energy usage, and latency while keeping security strong. It enables secure communication, and data integrity even on constrained hardware. Uses optimized block ciphers, stream ciphers and AEAD schemes. Example: Ascon-128a, chosen in NIST's lightweight crypto competition. It offers authenticated encryption and hashing with small footprint. Suitable for sensor networks, RFID tags, and embedded systems. Helps balance security and performance in IoT deployments.

40. List and briefly explain any three common IoT-specific attacks. What mitigation strategies can be applied?

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Ans: Three common IoT-specific attacks:

1. Firmware hijacking/malicious updates - attacker installs modified firmware.

→ Mitigation: Signed firmware, secure boot, rollback protection.

2. Physical tampering/debug-port abuse - attacker accesses internal memory or keys.

→ Mitigation: tamper-proof casing, disable JTAG/SWD, secure elements chips.

3. Botnets (e.g., Mirai) - exploit weak/default passwords, open Telnet/SSH.

→ Mitigation: unique strong credentials, disable unused services, firewall.

Other strategies: automatic patching, intrusion detection, network segmentation.

Goal: prevent device takeover, protect data, and ensure reliability.

Combines hardware, software and network defenses.